16-bit signed integer an internal software exception occurred. This caused the operand error, and caused the on-board computer to send, instead of real flight data, diagnostic data. This data erroneously commanded sudden nozzle deflections and also induced a high angle of attack

There were quite a few things to take away from this disaster. A clear simulation of the whole trajectory should have recognised immediately the overflow error and should have been performed at the very initial stages like the architectural design or the prototyping stage.

Data overflow software failure where the onboard computer receives only the diagnostic information instead of actual realistic flight data can be understood by viewing what happened to flight 501. There was a backup system of the malfunctioning part in the Ariane , but this was not able to help as it unfortunately used the same software code. Right now there are some notable points which were understood after this disaster

- A test facility should be prepared including as much real time equipment as logistics will allow, injecting realistic data, and most importantly conducting complete and closed-loop testing of the entire system. What also became clear is that although a subsystem can work perfectly on a previous embedded control system, it can still fail as a subsystem in a different new embedded system. In short, the reusability of a subsystem should always be questioned. In the case of flight 501, software for Ariane 5 was considered and trusted mainly because it ran smoothly on the Ariane 4, a clear mistake, as the static code analysis showed the overflow error.
- A sensor must return reliable data in all circumstances. Due to a data overflow of a 16 bit signed integer value the Inertial Reference System (IRS) returned diagnostic data instead of actual flight data. Unreliable data is worse than no data.
- This whole spectacle with the high risks associated with complex systems such as the Ariane 5 and embedded systems fortunately took the interest of politicians, executives, and the general public. This resulted in a large increase in support for research on ensuring the functionality and reliability of safety-critical systems. It is essential that there be be a trade off between safety, economic, and environmental aspects as these are all equally responsible and integral to creating a safe, sustainable, reliable system.